School of Electrical Engineering and Computer Sciences
National University of Sciences and Technology

DEPARTMENT OF Computing

Industry Advisory Board Meeting Minutes

Held on July 4th, 2023

In attendance

S.No.	Name	Designation/Organization/Company
1.	Dr. Muhammad Moazam Fraz (In-Chair)	Associate Professor, HoD AI and Data Science, Department of Computing, NUST-SEECS
2.	Dr. Muhaamad Imran Malik	Associate Professor, Head CS Department, NUST-SEECS
3.	Dr. Rafia Mumtaz	Professor, Head SE Department, NUST-SEECS
4.	Mr. Umer Nasim	Software Engineer/ Department Coordinator, NUST-SEECS
5.	Mr Shamyl Bin Mansoor	CTO, LearnOBots
6.	Mr. Syed Nauyan Rashid	Technical Lead (AI & ML) Redbuffer
7.	Muhammad Aasharib Nawshad	Deep Learning Engineer QLU.ai
8.	Asim Hameed Khan	System Analyst, Sky Electric

Venue: SEECS Seminar Hall / Online

Meeting time: 1500 hrs – 1600 hrs

Agenda Points / Discussion

Dr. Muhammad Moazam Fraz briefed the house about the department, the degree programs, the vision and mission of the department and how the curriculum is designed to achieve this vision and mission. He briefed the house about the inclusion of the new course in the undergraduate elective courses list named "Virtual and Augmented Reality" at NUST SEECS.

- Mr. Syed Nauyan commented that Including an undergraduate course on "Virtual and Augmented Reality" in BSCS, BSDS, BESE and BSAI programs at NUST SEECS is a strategic move that aligns perfectly with the demands of the job market in Pakistan. Virtual and augmented reality technologies are rapidly gaining momentum across various industries, including gaming, entertainment, education, healthcare, and manufacturing. By equipping students with the knowledge and skills in this field, NUST SEECS ensures that its graduates are well-prepared to tap into the immense opportunities arising from this growing market.

- He further added that professionals with expertise in virtual and augmented reality are in high demand, and companies are actively seeking individuals who can design and develop immersive experiences, create innovative solutions, and optimize user interactions. By offering a specialized course in this area, NUST SEECS not only enhances the employability of its graduates but also positions them for exciting and lucrative career prospects. The inclusion of this course enables students to stay ahead of the curve in an ever-evolving technological landscape, contributing to the growth and advancement of the digital ecosystem in Pakistan.

- Mr Aashariab added that the addition of the "Virtual and Augmented Reality" course in the BSCS, BSDS, BESE and BSAI programs at NUST SEECS offers significant advantages, especially in the context of its applicability in the development of immersive technologies for games. This course offers students a valuable opportunity to explore the rapidly evolving field of virtual and augmented reality, equipping them with the necessary knowledge and skills to create engaging and immersive game experiences. By delving into the principles of human response to stimuli and gaining proficiency in designing VR systems using technological building blocks, students are empowered to push the boundaries of game development. They can create interactive and realistic virtual environments, enhancing user experiences and opening new avenues for innovative gameplay mechanics. The course nurtures creativity, problem-solving, and technical expertise, providing a strong foundation for students aspiring to excel in the field of immersive game development. Ultimately, the inclusion of this course cultivates

a generation of skilled professionals capable of driving advancements in the ever-evolving landscape of virtual and augmented reality gaming.

- Mr Shamyl suggested that the addition of a course on "Virtual and Augmented Reality" in BSCS, BSDS, BSAI and BESE programs at NUST SEECS is a significant step towards enhancing education in the digital age. Virtual and Augmented Reality (VR/AR) technologies have revolutionized the way we interact with information and the world around us. By incorporating this course, students will gain valuable skills and knowledge that can be applied across various educational domains.

- He further added that VR/AR offers immersive and interactive experiences, enabling students to visualize complex concepts and theories, making them more engaging and memorable. It allows for realistic simulations and virtual laboratories, offering hands-on learning opportunities that are otherwise challenging to provide. Furthermore, the course will enable students to develop critical thinking and problem-solving skills as they design and create VR/AR applications. Incorporating VR/AR in education fosters creativity, collaboration, and innovation, preparing students for the technology-driven future. By offering this course, NUST SEECS acknowledges the importance of staying at the forefront of technological advancements and equipping students with the skills they need to thrive in the digital era.

- Mr Asim said I highly endorse the inclusion of an undergraduate course on "Virtual and Augmented Reality" in the BSCS, BSDS, and BSAI programs at NUST SEECS. Virtual and Augmented Reality (VR/AR) technologies have emerged as powerful tools with widespread applications in various industries. The entertainment industry, in particular, has witnessed a significant transformation with the integration of VR/AR, offering immersive experiences and captivating storytelling. By incorporating this course, NUST SEECS would equip its students with the necessary knowledge and skills to excel in this rapidly evolving field.

- He further added that VR/AR also plays a crucial role in data visualization, enabling users to interact and explore complex datasets in a more intuitive and immersive manner. In today's data-driven world, the ability to effectively visualize and analyze data is highly sought after by industries across domains. By training students in VR/AR data visualization techniques, NUST SEECS would empower them to become adept in transforming complex data into meaningful insights, thus addressing the growing demand for data professionals with advanced visualization skills.

- The inclusion of VR/AR in the curriculum presents exciting opportunities for students to explore freelance work and earn foreign remittances. With the increasing global demand for VR/AR content creation, freelance platforms provide a gateway for talented individuals to showcase their skills and collaborate with international clients. By encouraging students to explore freelancing opportunities in VR/AR, NUST SEECS can foster entrepreneurship and enhance their employability prospects, while also contributing to the country's foreign remittance earnings.

- Dr Moazam appreciated the feedback from the industry and mentioned that due efforts would be taken by the department to implement the suggested points.

DoC Department Coordinator
July 04, 2023

Mr Taufique ur Rehman
DLQEC SEECS
July 04, 2023